Inner Product space

Inner Product space: Let V be a real vector space. Suppose to each pair of vectors $u, v \in V$, there is assigned a real number, denoted by $\langle u, v\rangle$. This function is called a (real) inner product on V if it satisfies the following axioms:

- Linear Property: $\langle au_1 + bu_2, v\rangle = a\langle u_1, v\rangle + b\langle u_2, v\rangle$.
- Symmetric property: $\langle u, v\rangle = \langle v, u\rangle$.
- Positive Definite Property: $\langle u, u\rangle \geq 0$ and $\langle u, u\rangle = 0$ if and only if $u = 0$

The vector space V with an inner product is called a (real) inner product space.

Norm of a vector: The norm or length of a vector u, denoted by $\|u\|$ is given by $\|u\| = \sqrt{\langle u, u\rangle}$
By the third axiom of inner product, $\langle u, u\rangle$ is nonnegative for any vector u. Thus, its positive square root exists. The relation $\|u\|^2 = \langle u, u\rangle$ is used frequently.

Examples of Inner Product Spaces:

Example 1: Euclidian space R^n : Consider the vector space R^n. The dot product or scalar product in R^n is defined by
$u.v = a_1b_1 + a_2b_2 + \cdots + a_nb_n$
where $u = (a_1, a_2, \ldots, a_n)$ and $v = (b_1, b_2, \ldots, b_n)$. This function defines an inner product on R^n. The norm $\|u\|^2$ of the vector $u = (a_1, a_2, \ldots, a_n)$ in this space is as follows:
$\|u\|^2 = \sqrt{u.u} = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$

Example 2: Function space $C[a, b]$: The notation $C[a, b]$ is used to denote the vector space of all continuous functions on the closed interval $[a, b]$, where $a < t < b$. The following defines an inner product on $C[a, b]$, $f(t)$ and $g(t)$ in $C[a, b]$,
$\langle f, g\rangle = \int_a^b f(t)g(t)dt$. It is called the usual inner product on $C[a, b]$.
The vector space $P_n(t)$, of all polynomials is a subspace of $C[a, b]$ for any interval $[a, b]$ and hence, the above is also an inner product on $P_n(t)$.

Example 3: Matrix space $M = M_{m\times n}$: Let $M = M_{m\times n}$, the vector space of all real $m \times n$ matrices. An inner product is defined on M by
$\langle A, B\rangle = tr(B^T A)$
where, as usual, $tr()$ is the trace.
If $A = a_{ij}$ and $B = b_{ij}$, then $\langle A, B\rangle = tr(B^T A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}b_{ij}$ and $\|A\| = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$ That is, $\langle A, B\rangle$ is the sum of the products of the corresponding entries in A and B and, in particular, $\langle A, A\rangle$ is the sum of the squares of the entries of A.

Inner product spaces

Matrix space $M = M_{m,n}$

Let $M = M_{m,n}$, the vector space of all real $m \times n$ matrices. An inner product is defined on M by

$$\langle A, B \rangle = tr(B^T A)$$

Where $tr()$ is the sum of diagonal elements.
If $A = [a_{ij}]$ and $B = [b_{ij}]$, then

$$\langle A, B \rangle = tr(B^T A) = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ij}$$

and

$$\|A\|^2 = \langle A, A \rangle = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2$$

That is, $\langle A, B \rangle$ is the sum of the product of the corresponding entries in A and B and, in particular, $\langle A, A \rangle$ is the sum of the square of the entries of A.

Theorem 0.0.1 *Cauchy-Schwartz inequality*
For any vectors u and v in an inner product space V.

$$\langle u, v \rangle^2 \leq \langle u, u \rangle \langle v, v \rangle$$

or

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Angle between Vectors

For any non-zero vectors u and v in an inner product space V, the *angle* between u and v is defined to be the angle θ such that $0 \leq \theta \leq \pi$ and

$$cos\theta = \frac{\langle u, v \rangle}{\|u\|\|v\|}$$

By Cauchy-Schwartz inequality, $-1 \leq cos\theta \leq 1$, and so the angle exists and is unique.

Example 0.0.1 *a) Consider vectors* $u = (2, 3, 5)$ *and* $v = (1, -4, 3)$ *in* R^3. *Then,*
$\langle u, v \rangle = 2 - 12 + 15 = 5$
$\|u\| = \sqrt{4 + 9 + 25} = \sqrt{38}, \|v\| = \sqrt{1 + 16 + 9} = \sqrt{26}$
Then the angle θ *between* u *and* v *is given by*

$$cos\theta = \frac{5}{\sqrt{38}\sqrt{26}}$$

b) Let $f(t) = 3t - 5$ *and* $g(t) = t^2$ *in the polynomial space* $P(t)$ *with inner product* $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$.

Then $\langle f, g \rangle = \frac{-11}{12}$, $\|f\| = \sqrt{13}$, $\|g\| = \frac{1}{5}\sqrt{5}$
And angle between f *and* g *is given by*

$$cos\theta = \frac{-55}{12\sqrt{13}\sqrt{5}}$$

Orthogonality

Let V be an inner product space. Then vectors $u, v \in V$ are said to be *orthogonal* and u is said to be *orthogonal* to v if $\langle u, v \rangle = 0$
The relation is clearly symmetric, that is, if u is orthogonal to v, then $\langle v, u \rangle = 0$, and so v is orthogonal to u.

1 Orthogonal Complements:

Let S be a subset of an inner product space V. The orthogonal complement of S, denoted by $S^{\perp}$ consists of those vectors in V that are orthogonal to every $u \in S$; that is,

$$S^{\perp} = \{v \in V : \langle v, u\rangle = 0 \quad \text{for every} u \in S\}$$

In Particular, for a given vector u in V, we have

$$u^{\perp} = \{v \in V : \langle v, u\rangle = 0\}$$

that is $u^{\perp}$ consists of all vectors in V that are orthogonal to the given vector u. We show that $S^{\perp}$ is a subspace of V. Clearly $0 \in S^{\perp}$, since 0 is orthogonal to every vector in V. Now suppose $v, w \in S^{\perp}$. Then, for any scalars a and b and any vector $u \in S$, we have

$$\langle av + bw, u\rangle = a\langle v, u\rangle + b\langle w, u\rangle = a.0 + b.0 = 0$$

Thus $av + bw \in S^{\perp}$, and therefore $S^{\perp}$ is a subsapce of V. We stat this result formally.
Proposition: Let S be a subset of a vector space V. Then $S^{\perp}$ is a subspace of V.
Example: Find a basis for the subspace $u^{\perp}$ of R^3, where $u = (1, 3, -4)$. Note that $u^{\perp}$ consists of all vectors $w = (x, y, z)$ such that $\langle u, w\rangle = 0$, or $x + 3y - 4z = 0$. The free variables are y and z.
(1) Set $y = 1, z = 0$ to obtain the solution $w_1 = (-3, 1, 0)$
(2) Set $y = 0, z = 1$ to obtain the solution $W_1 = (4, 0, 1)$
The vectors w_1 and w_2 form a basis for the solution space of the equation, and hence a basis for $u^{\perp}$.
Suppose **W** is a subspace of V. Then both W and $W^{\perp}$ are subspaces of V. The next theorem, whose proof requires of later sections, is a basis result in linear algebra.
Theorem: Let W be a subspace of V. Then V is the direct sum of W and $W^{\perp}$, that is, $V = W \oplus W^{\perp}$.

Projections: Let V be an inner product space. Supplose w is a given nonzero vector in V, and suppose v is another vector. We seek the "Projection of v along w", will be the multiple cw of w such that $v' = v - cw$ is orthogonal to w. This means

$$\langle v - cw, w\rangle = 0 \quad \text{or} \quad \langle v, w\rangle - c\langle w, w\rangle = 0 \quad \text{or} \quad c = \frac{\langle v, w\rangle}{\langle w, w\rangle}.$$

Gram - schmidt orthogonalization process: Suppose $\{v_1, v_2, ..., v_n\}$ is a basis of an inner product space V. One can use this basis to construct an orthogonal basis $\{w_1, w_2, ..., w_n\}$ of V as follows, Set

$$w_1 = v_1$$

$$w_2 = v_2 - \frac{\langle v_2, w_1\rangle}{\langle w_1, w_1\rangle} w_1$$

$$w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

and so on

$$w_n = v_n - \frac{\langle v_n, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_n, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 - ... - \frac{\langle v_n, w_(n-1) \rangle}{\langle w_(n-1), w_(n-1) \rangle} w_(n-1).$$

1. Let V be the vector space of polyniomials $f(t)$ with inner product $< f, g > \int_{-1}^{1} f(t)g(t)dt$.Apply the Gram-Schmidt process to $(1, t, t^2, t^3)$ to find an orthogonal basis $\{f_0, f_1, f_2, f_3\}$ with integer coefficient for $P_3(t)$

Here we use the fact that, for $r + s = n$,

$< t^r, t^s >= \int_{-1}^{1} t^n dt = [\frac{t^{n+1}}{n+1}]_{-1}^{1} = \{\frac{2}{n+1}$ when n is even

and 0 when n is odd

(1) First set $f_0 = 1$

(2) Compute $t = \frac{<t,1>}{<1,1>}(1) = t - 0 = t$, set $f_1 = t$

(3) Compute

$t^2 - \frac{<t^2,1>}{<1,1>}(1) - \frac{<t^2,t>}{<t,t>}(t) = t^2 - \frac{\frac{2}{3}}{2}(1) + 0(t) = t^2 - \frac{1}{3}$

multiply by 3 to obtain $f_2 = 3t^2 = 1$.

(4) Compute

$t^3 - \frac{<t^3,1>}{<1,1>}(1) - \frac{<t^3,t>}{<t,t>}(t) = \frac{(t^3, 3t^2-1)}{(3t^2-1, 3t^2-1)}(3t^2 - 1)$

$= t^3 - 0(t) - \frac{\frac{2}{5}}{\frac{2}{3}}(t) - 0(3t^2 - 1) = t^3 - \frac{3}{5}t$

multiply by 5 to obtain $f_3 = 5t^2 - 3t$.

Thus $\{1, t, 3t^2 - 1, 5t^3 - 3t\}$ is the required orthogonal basis.

2. Find the $cos\theta$ where θ is the angle between:

(a) $u = (1, 3, -5, 4)$ and $v = 2, -3, 4, 1)$ in R^4

(b) $A = \begin{pmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \end{pmatrix}$

and $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$

where $(A, B) = tr(B^T A)$.

Use $cos\theta = \frac{<u,v>}{||u||||v||}$

(a) Compute:

$< u, v >= 2 - 9 - 20 + 4 = -23, \;\; ||u^2|| = 1 + 9 + 25 + 16 = 51, \;\; ||v^2|| = 4 + 9 + 16 + 1 = 30$

Thus $cos\theta = \frac{-23}{\sqrt{51}\sqrt{30}} = \frac{-23}{3\sqrt{170}}$

(b) Use $(A, B) = tr(B^T A) = \Sigma_{i=1}^{m}\Sigma_{j=1}^{n} a_{ij}b_{ij}$, the sum of the products of corresponding entries.

$(A, B) = 9 + 16 + 21 + 24 + 25 + 24 = 119$

Use $||A^2|| =< A, A >= \Sigma_{i=1}^{m}\Sigma_{j=1}^{n} a_{ij}^2$, the sum of the squares of all the elements of A.——

$||A^2|| =< A, A >= 9^2 + 8^2 + 7^2 + 6^2 + 5^2 + 4^2 = 271,$ and so $||A|| = \sqrt{271}$

$||B^2|| =< B, B >= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 = 91,$ and so $||B|| = \sqrt{91}$

Thus $cos\theta = \frac{119}{\sqrt{271}\sqrt{91}}$

3. Let W be the subspace of R^5 spanned by $u = (1, 2, 3, -1, 2)$ and $v = (2, 4, 7, 2, -1)$.Find a basis of the orthogonal complement of W.
we seek all vectors $w = (x, y, z, s, t)$ such that
$(w, u) = x + 2y + 3z - s + 2t = 0$
$(w, v) = 2x + 4y + 7z + 2s - t = 0$
Eliminating x from the second equation, we find the equivalent system
$x + 2y + 3z - s + 2t = 0$
$2x + 4y + 7z + 2s - t = 0$
The free variables are y, s and t. Therefore
(1) set $y = -1, s = 0, t = 0$ to obtain the solution $w_1 = (2, -1, 0, 0, 0)$
(2) set $y = 0, s = 1, t = 0$ to obtain the solution $w_2 = (13, 0, -4, 1, 0)$
(3) set $y = 0, s = 0, t = 1$ to obtain the solution $w_3 = (-17, 0, 5, 0, 1)$
The set $[w_1, w_2, w_3]$ is a basis of orthogonal complement of W

4. Let $W = (1, 2, 3, 1)$ be a vector in R^4. Find the orthogonal basis for orthogonal comlement of W.
Find a nonzero solution of $x + 2y + 3z + t = 0$ say $v_1 = (0, 0, 1, -3)$. Now find a nonzero solution of the system
$x + 2y + 3z + t = 0,\ z - 3t = 0$
say $v_2 = (0, -5, 3, 1)$. Lastly, find a nonzero solution of the system
$x + 2y + 3z + t = 0,\ -5y + 3z + t = 0,\ z - 3t = 0$
say $v_3 = (-14, 2, 3, 1)$. Thus v_1, v_2, v_3 form an orthogonal basis for orthogonal comlement of W.

5. Consider the vector space $P(t)$ with inner product $< f, g >= \int_0^1 f(t)g(t)dt$. Apply the Gram-Schmidt algorithm to the set $\{1, t, t^2\}$ to obtain an orthogonal set $\{f_0, f_1, f_2\}$ with integer coefficients.
first set $f_0 = 1$, Then find
$t - \frac{<t,1>}{<1,1>}.1 = t - \frac{\frac{1}{2}}{1}.1 = t - \frac{1}{2}$
clear fractions to obtain $f_1 = 2t - 1$, Then find $t^2 - \frac{<t^2,1>}{<1,1>}(1) - \frac{<t^2,2t-1>}{<2t-1,2t-1>}(2t - 1) = t^2 - \frac{\frac{1}{3}}{1}(1) - = t - \frac{\frac{1}{6}}{\frac{1}{3}}(2t - 1) = t^2 - t + \frac{1}{6}$.
Clear fractions to obtain $f_2 = 6t^2 - 6t + 1$, Thus $\{1, 2t - 1, 6t^2 - 6t + 1\}$ is the required orthogonal set.